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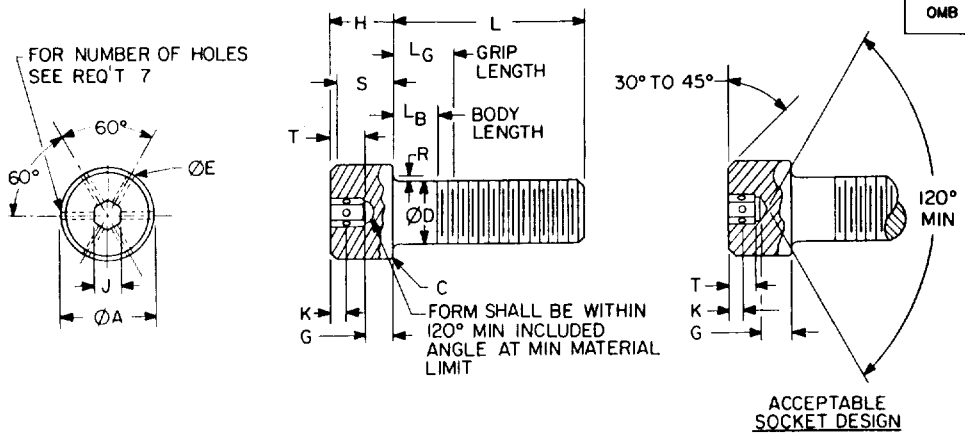


TABLE I. DASH NUMBERS AND DIMENSIONS

NOMINAL SIZE			#6 (.138)	#8 (.164)	#10 (.190)	1/4 (.250)
THREAD SIZE			.138-40UNF-3A	.164-36UNF-3A	.190-32UNF-3A	.250-28UNF-3A
ØD	BODY	MAX	.1380	.1640	.1900	.2500
		MIN	.1329	.1585	.1840	.2435
ØA	HEAD	MAX	.226	.270	.312	.375
		MIN	.218	.262	.303	.365
H	HEAD HEIGHT	MAX	.138	.164	.190	.250
		MIN	.134	.159	.185	.244
S	HEAD SIDE HEIGHT	MIN	.124	.148	.171	.225
J	SOCKET WIDTH ACROSS FLATS	MAX	.1111	.1426	.1587	.1900
		MIN	.1094	.1406	.1562	.1875
T	KEY ENGAGEMENT	MIN	.064	.077	.090	.120
G	WALL THICKNESS	MIN	.047	.056	.065	.095
C	CHAMFER OR RADIUS	MAX	.005	.005	.005	.008
R	FILLET	MAX	.010	.012	.014	.014
	EXTENSION	MIN	.006	.007	.009	.009
K	HOLE LOCATION	MAX	.050	.060	.065	.085
		MIN	.035	.040	.045	.065
ØE	HOLE	MAX	.039	.050	.050	.050
		MIN	.033	.044	.044	.044
	RECOMMENDED WIRE DIAMETER	MAX	.028	.040	.040	.040
		MIN	.020	.032	.032	.032
	ALIGNMENT PLUG DIAMETER 1/		.025	.030	.030	.030
	TENSILE STRENGTH, LBS 2/	MIN	1,830	2,650	3,600	6,190

L LENGTH	TOLERANCE	#6 (.138) DASH NO	#6 (.138) L_G MAX	#6 (.138) L_B MIN	#8 (.164) DASH NO	#8 (.164) L_G MAX	#8 (.164) L_B MIN	#10 (.190) DASH NO	#10 (.190) L_G MAX	#10 (.190) L_B MIN	1/4 (.250) DASH NO	1/4 (.250) L_G MAX	1/4 (.250) L_B MIN
.250	+0.00 -0.03	1									19		
.375		2			6			9			20		
.500		3			7			10			21		
.625		4						11			22		
.750		5			8			12			23		
.875								13			24		
1.000								14			25		
1.250	+0.00 -0.04							15	.38	.17	26	.50	.25
1.500								16	.38	.17	27	.50	.25
1.750								17	.88	.67	28	1.00	.75
2.000								18	.88	.67	29	1.00	.75
2.250													

1/ SEE REQUIREMENT 8

2/ SEE REQUIREMENT 9

(E) ENTIRE STANDARD REVISED

INCH - POUND

PREPARING ACTIVITY: ARMY-AR CUSTODIANS: ARMY- NAVY- OS AIR FORCE- 99 DLA- REVIEW: EA,ER,IS,SH USER: AS,AT,AV,MC,ME,YD PROJECT NUMBER: 5305-1917	MILITARY SPECIFICATION SHEET TITLE SCREW, CAP, SOCKET HEAD, HEXAGON, DRILLED ALLOY STEEL, (UNF-3A)	SPECIFICATION SHEET NUMBER MS24678E 2 JAN 92 SUPERSEDING MS24678D 5 MARCH 1986 AND MS24675 AMSC- N/A FSC - 5305
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TABLE I. DASH NUMBERS AND DIMENSIONS - CONTINUED

NOMINAL SIZE			5/16 (.3125)	3/8 (.375)	7/16 (.4375) 3/	1/2 (.500)
THREAD SIZE			.3125-24UNF-3A	.375-24UNF-3A	.4375-20UNF-3A	.500-20UNF-3A
ØD	BODY	MAX	.3125	.3750	.4375	.5000
		MIN	.3053	.3678	.4294	.4919
ØA	HEAD	MAX	.469	.562	.656	.750
		MIN	.457	.550	.642	.735
H	HEAD HEIGHT	MAX	.312	.375	.438	.500
		MIN	.306	.368	.430	.492
S	HEAD SIDE HEIGHT	MIN	.281	.337	.394	.450
J	SOCKET WIDTH ACROSS FLATS	MAX	.2530	.3160	.3790	.3790
		MIN	.2500	.3125	.3750	.3750
T	KEY ENGAGEMENT	MIN	.151	.182	.213	.245
G	WALL THICKNESS	MIN	.119	.143	.166	.190
C	CHAMFER OR RADIUS	MAX	.008	.008	.010	.010
R	FILLET EXTENSION	MAX	.017	.020	.023	.026
		MIN	.012	.015	.018	.020
K	HOLE LOCATION	MAX	.104	.123	.141	.160
		MIN	.084	.103	.121	.140
ØE	HOLE	MAX	.050	.067	.067	.067
		MIN	.044	.061	.061	.061
	RECOMMENDED WIRE DIAMETER	MAX	.040	.057	.057	.057
		MIN	.032	.045	.045	.045
	ALIGNMENT PLUG DIAMETER 1/		.030	.051	.051	.051
	TENSILE STRENGTH, LBS 2/	MIN	9,860	14,900	20,200	27,200

L LENGTH	TOLERANCE SIZE .375 OR UNDER	TOLERANCE SIZE .4375 AND OVER	DASH NO	L_G MAX	L_B MIN	DASH NO	L_G MAX	L_B MIN	DASH NO	L_G MAX	L_B MIN	DASH NO	L_G MAX	L_B MIN
.500 .625 .750 .875 1.000	+0.00 -0.03	+0.00 -0.03	30 31 32 33 34 35 36			41 42 43 44 45 46 47			53 54 55			57 58 59 60 61		
1.250 1.500 1.750 2.000 2.250 2.500	+0.00 -0.04	+0.00 -0.06	37 38 39 40	.62 .62 1.12 1.12	.35 .35 .85 .85	48 49 50 51 52	.50 .50 1.00 1.00 1.50	.19 .19 .69 .69 1.19	56	.62	.27	62 63 64	.75 .75 1.50	.36 .36 1.12
3.000	+0.00 -0.06	+0.00 -0.08												

1/ SEE REQUIREMENT 8

2/ SEE REQUIREMENT 9

3/ INACTIVE FOR NEW DESIGN

PREPARING ACTIVITY: ARMY-AR CUSTODIANS: ARMY- NAVY- OS AIR FORCE- 99 DLA- REVIEW: EA,ER,IS,SH USER: AS,AT,AV,MC,ME,YD PROJECT NUMBER: 5305-1917	MILITARY SPECIFICATION SHEET TITLE SCREW, CAP, SOCKET HEAD, HEXAGON, DRILLED ALLOY STEEL, (UNF-3A)	SPECIFICATION SHEET NUMBER MS24678E 2 JAN 92 SUPERSEDING MS24678D 5 MARCH 1986 AND MS24675 AMSC- N/A FSC - 5305
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TABLE I. DASH NUMBERS AND DIMENSIONS - CONTINUED

NOMINAL SIZE			5/8 (.625)
THREAD SIZE			.625-18UNF-3A
ØD	BODY	MAX MIN	.6250 .6163
ØA	HEAD	MAX MIN	.938 .921
H	HEAD HEIGHT	MAX MIN	.625 .616
S	HEAD SIDE HEIGHT	MIN	.562
J	SOCKET WIDTH ACROSS FLATS	MAX MIN	.5050 .5000
T	KEY ENGAGEMENT	MIN	.307
G	WALL THICKNESS	MIN	.238
C	CHAMFER OR RADIUS	MAX	.010
R	FILLET EXTENSION	MAX MIN	.032 .024
K	HOLE LOCATION	MAX MIN	.198 .178
ØE	HOLE	MAX MIN	.067 .061
	RECOMMENDED WIRE	MAX MIN	.057 .045
1/	ALIGNMENT PLUG DIAMETER		.051
2/	TENSILE STRENGTH, LBS	MIN	43,500

L LENGTH	TOLERANCE	DASH NO	L_G MAX	L_B MIN
1.000	+0.00 -0.03	65		
1.250		66		
1.500		67		
1.750	+0.00	68		
2.000	-0.06	69		
2.500		70	.75	.30

1/ SEE REQUIREMENT 8

2/ SEE REQUIREMENT 9

INTERCHANGEABILITY DATA

EFFECTIVE 7 JULY 1975, MS24675 WAS CANCELLED AND SUPERSEDED BY MS24678

MS24675 AND THE SCREWS COVERED BY THE PART NUMBERS LISTED IN THE STANDARD ARE CANCELLED AFTER THE DATE INDICATED ON THIS DOCUMENT.

USE ONLY THE PART NUMBER FROM MS24678 FOR DESIGN AND REPLACEMENT.

CANCELLED MS24675 SCREWS CANNOT ALWAYS REPLACE MS24578 SCREWS AND SHOULD BE USED ONLY WHERE APPLICABLE UNTIL EXISTING STOCK IS DEPLETED.

MS24678 SCREWS SHALL BE USED IN LIEU OF THE CANCELLED MS24675 SCREWS OF THE SAME DASH NUMBERS. FOR EXAMPLE:

SCREWS MS24675-1 THROUGH MS24675-70
ARE CANCELLED AND SUPERSEDED BY
SCREWS MS24678-1 THROUGH MS24678-70

PREPARING ACTIVITY: ARMY-AR
CUSTODIANS: ARMY- NAVY- OS
AIR FORCE- 99 DLA-
REVIEW: EA,ER,IS,SH
USER: AS,AT,AV,MC,ME,YD
PROJECT NUMBER: 5305-1917

MILITARY SPECIFICATION SHEET

TITLE
SCREW, CAP, SOCKET HEAD, HEXAGON, DRILLED ALLOY STEEL, (UNF-3A)

SPECIFICATION SHEET NUMBER
MS24678E 2 JAN 92
SUPERSEDING MS24678D 5 MARCH 1986 AND MS24675
AMSC- N/A FSC - 5305

DISTRIBUTION STATEMENT
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REQUIREMENTS:

1. MATERIAL: ALLOY STEEL IN ACCORDANCE WITH PROCUREMENT SPECIFICATION.

2. PROTECTIVE COATING AND SURFACE TREATMENT: UNCOATED, CADMIUM PLATE IN ACCORDANCE WITH QQ-P-416, TYPE II, CLASS 3, OR ALUMINUM ION VAPOR DEPOSITED IN ACCORDANCE WITH MIL-C-83488, TYPE II, CLASS 3 (SEE REQT 11).

   (a) DO NOT USE CADMIUM PLATED SCREWS FOR TEMPERATURES ABOVE 450°.

   (b) ALUMINUM ION VAPOR DEPOSITED COATING CAN BE USED FOR TEMPERATURES UP TO 850°F.

3. HARDNESS: IN ACCORDANCE WITH PROCUREMENT SPECIFICATION.

4. THREADS: THREADS SHALL BE IN ACCORDANCE WITH FED-STD-H28/2. ACCEPTABILITY OF SCREW THREADS SHALL BE IN ACCORDANCE WITH FED-STD-H28/20, SYSTEM 22.

5. THREAD LENGTH: SCREWS ABOVE HEAVY LINE SHALL HAVE COMPLETE (FULL FORM) THREADS EXTENDING WITHIN TWO THREADS OF THE HEAD.

6. HEAD: HEAD MAY OR MAY NOT BE KNURLED AT MANUFACTURER'S OPTION. TYPE OF KNURLING OPTIONAL. DIAMETER OF KNURLED HEAD SHALL NOT EXCEED THE MAXIMUM HEAD DIAMETER SPECIFIED.

7. DRILLED HOLE DATA:

   (a) SCREWS #6 (.138) SHALL HAVE ONE (1) HOLE DRILLED THROUGH THE HEAD. SCREWS #8 (.164) AND LARGER SHALL HAVE (3) HOLES DRILLED THROUGH THE HEAD.

   (b) ON SCREWS SMALLER THAN #8 (.164) THE DRILLED HOLE SHALL BE CENTERED AS CLOSE AS PRACTICABLE ON THE FLAT OF THE HEXAGON SOCKET. ON SCREWS #8 (.164) AND LARGER THE DRILLED HOLES SHALL BE WITHIN THE FLATS ON THE HEXAGON SOCKET AND SHALL NOT BREAK THE CORNERS OF THE HEXAGON SOCKET.

   (c) EDGES OF THE HOLES ON THE OUTSIDE OF THE HEAD SHALL BE CHAMFERED 45° BY 0.010 TO 0.020 DEEP.

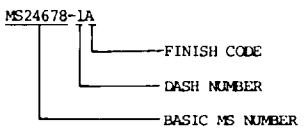
8. ALIGNMENT PLUG: PLUG SHALL PASS COMPLETELY THROUGH THE DRILLED HOLES IN THE HEAD WITHOUT DEFLECTION.

9. TENSILE STRENGTH: BASED ON MINIMUM ULTIMATE TENSILE OF 180.000 PSI FOR SIZES #4 (.112) THRU #10 (.190) AND 170,000 PSI FOR SIZES OVER #10 (.190). LOAD POUNDS CALCULATED BY THE STRESS AREAS INDICATED IN FED-STD-H28/2.

10. FINISH CODE:

    A - ALUMINUM COATING
    U - NO PROTECTIVE COATING OR SURFACE TREATMENT
    NO CODE - CADMIUM PLATE (NOTE: MS24678D PART NUMBERS IN GOVERNMENT INVENTORY WITH ADDED "C" SHALL BE USED UNTIL STOCK IS DEPLETED.)

11. PART NUMBER: THE PART NUMBER SHALL CONSIST OF THE BASIC MS NUMBER FOLLOWED BY A DASH NUMBER TAKEN FROM TABLE I, FOLLOWED BY A FINISH CODE LETTER, IF APPLICABLE.



MS24678-1A INDICATES - SCREW, CAP, SOCKET HEAD, HEXAGON, DRILLED ALLOY STEEL; .138-40 UNF-3A; LENGTH .250; ALUMINUM COATING.

MS24678-1 INDICATES - SCREW, CAP, SOCKET HEAD, HEXAGON, DRILLED ALLOY STEEL; .138-40 UNF-3A; LENGTH .250; CADMIUM PLATE.

MS24678-1U INDICATES - SCREW, CAP, SOCKET HEAD, HEXAGON, DRILLED ALLOY STEEL, .138-40 UNC-3A, LENGTH .250; NO PROTECTIVE COATING OR SURFACE TREATMENT.

12. SOURCE IDENTIFICATION MARK: SCREWS WITH NOMINAL SIZES .190 AND LARGER SHALL BE PERMANENTLY MARKED TO IDENTIFY THE SOURCE ACCEPTING RESPONSIBILITY FOR THE SCREWS MEETING THE REQUIREMENTS SPECIFIED HEREIN. THE MARKING SHALL BE A SOURCE IDENTIFYING SYMBOL FOR A MANUFACTURER IN ACCORDANCE WITH GOVERNMENT REGULATIONS OR A PRIVATE LABEL DISTRIBUTOR'S SYMBOL AS APPLICABLE.

NOTES:

1. ALL DIMENSIONS ARE IN INCHES.

2. IN THE EVENT OF A CONFLICT BETWEEN THE TEXT OF THIS STANDARD AND THE REFERENCES CITED HEREIN, THE TEXT OF THIS STANDARD SHALL TAKE PRECEDENCE.

3. REFERENCED GOVERNMENT (OR NON-GOVERNMENT) DOCUMENTS OF THE ISSUE LISTED IN THAT ISSUE OF THE DEPARAMENT OF DEFENSE INDEX OF SPECIFICATIONS AND STANDARDS (DODISS) SPECIFIED IN THE SOLICITATION FORM A PART OF THIS STANDARD TO THE EXTENT SPECIFIED HEREIN.

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